Week 06 teacher guide

Expressions and equations

PowerPoint slides 486-576 | Student workbook pp. 46-53

Expressions name quantities. Equivalent forms describe the same quantity. Equations can be solved by keeping both sides equal.

The seven days

Day 036 · p. 46 · Saturday · 20 questions,including 2 challenges
Expressions and equivalent forms

Day 037 · pp. 47-48 · Sunday · 20 questions,including 2 challenges
Equations and related quantities

Day 038 · p. 49 · Monday · 20 questions,including 2 challenges
Expressions and equations

Day 039 · p. 50 · Tuesday · 20 questions,including 2 challenges
Expressions and equations

Day 040 · p. 51 · Wednesday · 25 questions,including 2 challenges
Expressions and equations

Day 041 · p. 52 · Thursday · 20 questions,including 2 challenges
Expressions and equations

Day 042 · p. 53 · Friday · 20 questions,including 2 challenges
Weekly check

Use one day at a time

The student pages contain questions only. Work on separate plain paper, labeling the Day number and item IDs. Weekend lessons use 20 core questions including C1 and C2. Weekdays use 20 questions; Wednesday uses 25. Allow 30-45 minutes on ordinary weekdays and 40-55 minutes on Wednesday. Split a demanding day into two sittings if needed.

The Sunday reference page is for checking a method after an attempt. Keep answer-review slides out of view until the corresponding work is complete. Weekly answer-review sections are optional after Friday and do not add to the 80-minute weekend lesson.

Built-in review

Day 038: Expressions and equations (9, 10, 11, 12)

Day 039: Expressions and equations (9, 10, 11, 12)

Day 040: Weekly puzzle (13)

Day 041: Decimal practice (9, 10, 11, 12)

On Friday, record routine accuracy out of 18 separately from the two challenge explanations. For each challenge, note whether the learner connected the steps, used a valid model or calculation, and justified the conclusion. Revisit a missed model, then check with a fresh item later.

Saturday. Expressions and equivalent forms

Day 036 · p. 46 | Slides 486-515

Goal: Write expressions and show why two forms have the same value.

Check first: Multiplication, fraction products, and grouping symbols.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 486-487 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 488-489 · Recall · R1, R2, R3, R4

Try from memory, then check. Use one error for a quick reminder; keep this brief.

Slides 490-491 · Explore · P1, P2, P3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 492 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 493-494 · Explore · P4, P5, P6

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 495-497 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 498 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 499 · Compare methods

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 500 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 501 · Language

Attach each term to a model. Ask for an example and a non-example.

Slides 502-505 · Guided practice · X1, X2, X3, X4

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 506-507 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 508-511 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 512 · Summary · S

Ask for one sentence or annotated example in the learner’s own words. Listen for the main relationship.

Slides 513 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 514 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 515 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Saturday. Discussion and support

Investigation T: One rectangle, three expressions

Watch and discuss; this is a projected investigation, not an extra assigned question.

A rectangle has height 4 and width 2x + 3. Ari writes 4(2x + 3), Bo writes 8x + 12, and Cy writes 8x + 3. Decide which expressions match for every x. Use a split rectangle, then test x = 2. Explain why testing one value can disprove a claim but cannot establish every-value equivalence.

1. Split the width into 2x and 3; both parts have height 4.

2. Their areas are 8x and 12, so 4(2x + 3) = 8x + 12.

3. At x = 2, Ari and Bo both give 28; Cy gives 19. A structural split explains every-value equivalence.

Ask: Which part of the rectangle accounts for the constant 12?

If stuck: Trace the height 4 on both parts; do not add the labels as lengths from different directions.

Expected: Ari and Bo are equivalent; Cy is not. At x = 2 the values are 28, 28, 19.

C1 · Challenge discussion

Ask: What does each 6 multiply, and why?

If stuck: Draw six boxes; write x and 4 in every box.

Expected: 6(x + 4) = 6x + 24; 66 tools.

C2 · Challenge discussion

Ask: Which lengths add when the banners are placed side by side?

If stuck: Keep the height 3; combine only the widths.

Expected: 6x + 6 square m; 30 square m; one 3 m by (2x + 2) m rectangle.

What to notice

Draws repeated groups before writing 5n. → Ask what changes when one sheet costs more.

Partitions the rectangle and counts both areas. → Ask which area matches each term.

Common error: Writes $4^2 = 8$. Repair: Write two factors of 4, then compare.

Sunday. Equations and related quantities

Day 037 · pp. 47-48 | Slides 516-543

Goal: Solve equations by keeping both sides equal and connect a rule to a table and graph.

Check first: Equivalent expressions and inverse operations.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 516-517 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 518-519 · Recall · R1, R2, R3, R4

Try from memory, then check. Use one error for a quick reminder; keep this brief.

Slides 520-521 · Explore · P1, P2, P3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 522 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 523-524 · Explore · P4, P5, P6

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 525-527 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 528 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 529 · Compare methods

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 530 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 531 · Language

Attach each term to a model. Ask for an example and a non-example.

Slides 532 · Explore · M1, M2, M3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 533 · Guided practice · X1

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 534-535 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 536-539 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 540 · Summary · S

Ask for one sentence or annotated example in the learner's own words. Listen for the main relationship.

Slides 541 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 542 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 543 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Sunday. Discussion and support

Investigation T: A solution and a changing relationship

Watch and discuss; this is a projected investigation, not an extra assigned question.

A pool contains 6 liters and gains 3 liters each minute. Compare the equation $6 + 3t = 24$ with the rule $L = 6 + 3t$. Solve the equation, build rule outputs for $t = 0, 2, 6$, and explain why the rule has many pairs but the equation asks for one time.

1. At zero minutes the pool already has 6 liters.
2. The rule $L = 6 + 3t$ gives 6, 12, 24 liters at 0, 2, 6 minutes.
3. Set $L = 24$: remove 6, then divide 18 by 3 to get 6 minutes.

Ask: Which quantity is fixed in the equation and which may vary in the rule?

If stuck: Cover L with 24 to turn the rule into the one-question equation.

Expected: $t = 6$; rule pairs (0, 6), (2, 12), (6, 24).

C1 · Challenge discussion

Ask: Which part of the total is not repeated for every bag?

If stuck: Separate the one-time fee from the equal bag costs.

Expected: $5 per bag; first shop is cheaper by $1 for five bags.

C2 · Challenge discussion

Ask: What must the rule give when the time is zero?

If stuck: Fill the zero-minute row first.

Expected: Table: (0, 3), (1, 5), (3, 9), (5, 13); $L = 3 + 2t$; 7 minutes.

What to notice

Removes or divides equally on both sides. → Ask the student to state why the balance remains level.

Splits the fractional whole into equal parts. → Ask which parts the given amount represents.

Common error: Subtracts the coefficient. Repair: Read 4b as four equal groups of b.

Slide-order index

Follow this numerical order. The session plans explain how to teach each phase; full scripts remain in PowerPoint speaker notes. Item IDs match the student workbook.

Saturday

486. Week 6, Saturday
Watch and discuss

487. Today
Watch and discuss

488. Recall
Day 036 · Workbook p. 46 · R1, R2, R3, R4

489. Recall: answers
Day 036 · Workbook p. 46 · R1, R2, R3, R4

490. Expression from a story
Day 036 · Workbook p. 46 · P1

491. Evaluate using order of operations
Day 036 · Workbook p. 46 · P2, P3

492. The model
Watch and discuss

493. Distributive property from area
Day 036 · Workbook p. 46 · P4

494. Combine like terms
Day 036 · Workbook p. 46 · P5, P6

495. Worked example · 1 of 3
Watch and discuss · worked example

496. Worked example · 2 of 3
Watch and discuss · worked example

497. Worked example · 3 of 3
Watch and discuss · worked example

498. Your turn
Day 036 · Workbook p. 46 · F

499. Two ways to see it
Watch and discuss · compare the same example

500. A claim to check
Day 036 · Workbook p. 46 · D

501. Words and meaning
Watch and discuss

502. Guided practice · X1
Day 036 · Workbook p. 46 · X1

503. Guided practice · X2
Day 036 · Workbook p. 46 · X2

504. Guided practice · X3
Day 036 · Workbook p. 46 · X3

505. Guided practice · X4
Day 036 · Workbook p. 46 · X4

506. One rectangle, three expressions
Watch and discuss · draw on separate paper

507. One rectangle, three expressions
Watch and discuss · draw on separate paper

508. Challenge C1
Day 036 · Workbook p. 46 · C1

509. Challenge C1 · Discuss
Day 036 · Workbook p. 46 · C1

510. Challenge C2
Day 036 · Workbook p. 46 · C2

511. Challenge C2 · Discuss
Day 036 · Workbook p. 46 · C2

512. In your own words
Day 036 · Workbook p. 46 · S

513. Exit ticket
Day 036 · Workbook p. 46 · E

514. The week ahead
Days 038-042 · Workbook pp. 49-53

515. Tutor reference
Tutor only;hidden in slide show

Sunday

516. Week 6, Sunday
Watch and discuss

517. Today
Watch and discuss

518. Recall
Day 037 · Workbook p. 47 · R1, R2, R3, R4

519. Recall: answers
Day 037 · Workbook p. 47 · R1, R2, R3, R4

520. Additive equation with whole numbers
Day 037 · Workbook p. 47 · P1

521. Multiplicative equation with whole numbers
Day 037 · Workbook p. 47 · P2, P3

522. The model
Watch and discuss

523. Fractional coefficient
Day 037 · Workbook p. 47 · P4, P5

524. Equation
Day 037 · Workbook p. 47 · P6

525. Worked example · 1 of 3
Watch and discuss · worked example

526. Worked example · 2 of 3
Watch and discuss · worked example

527. Worked example · 3 of 3
Watch and discuss · worked example

528. Your turn
Day 037 · Workbook p. 47 · F

529. Two ways to see it
Watch and discuss · compare the same example

530. A claim to check
Day 037 · Workbook p. 47 · D

531. Words and meaning
Watch and discuss

532. Mixed review
Day 037 · Workbook p. 47 · M1, M2, M3

533. Guided practice · X1
Day 037 · Workbook p. 47 · X1

534. A solution and a changing relationship
Watch and discuss · draw on separate paper

535. A solution and a changing relationship
Watch and discuss · draw on separate paper

536. Challenge C1
Day 037 · Workbook p. 47 · C1

537. Challenge C1 · Discuss
Day 037 · Workbook p. 47 · C1

538. Challenge C2
Day 037 · Workbook p. 47 · C2

539. Challenge C2 · Discuss
Day 037 · Workbook p. 47 · C2

540. In your own words
Day 037 · Workbook p. 47 · S

541. Exit ticket
Day 037 · Workbook p. 47 · E

542. The week ahead
Days 038-042 · Workbook pp. 49-53

543. Tutor reference
Tutor only;hidden in slide show

Answer review

544. Week 6 answers
Next lesson: slide 577. Use this review section after Friday.

545. Expressions and equivalent forms · 1 of 5
Day 036 · Workbook p. 46 · Items R1, R2, R3, R4, P1

546. Expressions and equivalent forms · 2 of 5
Day 036 · Workbook p. 46 · Items P2, P3, P4, P5, P6

547. Expressions and equivalent forms · 3 of 5
Day 036 · Workbook p. 46 · Items F, D, X1, X2, X3

548. Expressions and equivalent forms · 4 of 5
Day 036 · Workbook p. 46 · Items X4, S, E, C1

549. Expressions and equivalent forms · 5 of 5
Day 036 · Workbook p. 46 · Items C2

550. Equations and related quantities · 1 of 5
Day 037 · Workbook p. 47 · Items R1, R2, R3, R4, P1

551. Equations and related quantities · 2 of 5
Day 037 · Workbook p. 47 · Items P2, P3, P4, P5, P6

552. Equations and related quantities · 3 of 5
Day 037 · Workbook p. 47 · Items F, D, M1, M2, M3

553. Equations and related quantities · 4 of 5
Day 037 · Workbook p. 47 · Items X1, S, E, C1

554. Equations and related quantities · 5 of 5
Day 037 · Workbook p. 47 · Items C2

555. Expressions and equations · 1 of 4
Day 038 · Workbook p. 49 · Items 1, 2, 3, 4, 5

556. Expressions and equations · 2 of 4
Day 038 · Workbook p. 49 · Items 6, 7, 8, 9, 10

557. Expressions and equations · 3 of 4
Day 038 · Workbook p. 49 · Items 11, 12, 13, 14, 15

558. Expressions and equations · 4 of 4
Day 038 · Workbook p. 49 · Items 16, 17, 18, C1, C2

559. Expressions and equations · 1 of 4
Day 039 · Workbook p. 50 · Items 1, 2, 3, 4, 5

560. Expressions and equations · 2 of 4
Day 039 · Workbook p. 50 · Items 6, 7, 8, 9, 10

561. Expressions and equations · 3 of 4
Day 039 · Workbook p. 50 · Items 11, 12, 13, 14, 15

562. Expressions and equations · 4 of 4
Day 039 · Workbook p. 50 · Items 16, 17, 18, C1, C2

563. Expressions and equations · 1 of 5
Day 040 · Workbook p. 51 · Items 1, 2, 3, 4, 5

564. Expressions and equations · 2 of 5
Day 040 · Workbook p. 51 · Items 6, 7, 8, 9, 10

565. Expressions and equations · 3 of 5
Day 040 · Workbook p. 51 · Items 11, 12, 13, 14, 15

566. Expressions and equations · 4 of 5
Day 040 · Workbook p. 51 · Items 16, 17, 18, 19, 20

567. Expressions and equations · 5 of 5
Day 040 · Workbook p. 51 · Items 21, 22, 23, C1, C2

568. Expressions and equations · 1 of 5
Day 041 · Workbook p. 52 · Items 1, 2, 3, 4, 5

569. Expressions and equations · 2 of 5
Day 041 · Workbook p. 52 · Items 6, 7, 8, 9, 10

570. Expressions and equations · 3 of 5
Day 041 · Workbook p. 52 · Items 11, 12, 13, 14, 15

571. Expressions and equations · 4 of 5
Day 041 · Workbook p. 52 · Items 16, 17, 18, C1

572. Expressions and equations · 5 of 5
Day 041 · Workbook p. 52 · Items C2

573. Weekly check · 1 of 4
Day 042 · Workbook p. 53 · Items 1, 2, 3, 4, 5

574. Weekly check · 2 of 4
Day 042 · Workbook p. 53 · Items 6, 7, 8, 9, 10

575. Weekly check · 3 of 4
Day 042 · Workbook p. 53 · Items 11, 12, 13, 14, 15

576. Weekly check · 4 of 4
Day 042 · Workbook p. 53 · Items 16, 17, 18, C1, C2

Complete weekly answers

Match Day,item ID,and page before checking. Equivalent correct forms and valid alternative methods are welcome. Computational keys are compact; challenges and conceptual items include reasoning. No challenge is complete with an unsupported answer alone.

Day 036 · Expressions and equivalent forms

Workbook p. 46 · assigned set

R1 · R-RATE

0.65 dollars per marker

R2 · F-DIV

6

R3 · N-DIV

12

R4 · P-WHOLE

60 shells

P1 · E-EXPR

5n; $10; $20

P2 · E-EVAL

17

P3 · E-EXP

16; 8; 27; $\frac{4}{9}$. 4^2 and 4×2 have different meanings.

Reason: Write repeated factors: 4×4, 4×2, $3 \times 3 \times 3$, and $\frac{2}{3} \times \frac{2}{3}$.

P4 · E-DIST

$4(x + 3) = 4x + 12$ square units

P5 · E-LIKE

5x; 15 when x = 3

Reason: There are five groups of x.

P6 · E-DIST

No: $12 \neq 6$. The equivalent form is $4t + 8$.

Reason: One counterexample proves the forms differ. Distribution gives $4t + 8$.

F · E-DIST

12; both forms give 22

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · E-DIST

$5z + 10$. Each of the five groups contains two extra units.

Reason: Lena multiplied one part only. Draw five equal rows, each with z and 2.

X1 · E-EXPR

$3p + 2$; 8 cards.

X2 · E-EVAL

7

Reason: $2 \times 2 + 3 = 4 + 3 = 7$.

X3 · E-EXP

7

Reason: The exponent gives two equal factors: 2×2; add 3 afterward.

X4 · E-DIST

$2x + 6$; both equal 10.

Reason: Multiply both x and 3 by 2.

S · EXPLAIN

Answers vary. An expression names a quantity. Equivalent expressions name the same quantity for every allowed value.

Reason: Accept an accurate explanation supported by a matching example.

E · E-DIST

$6x + 12$

Reason: $3 \times 2x = 6x$ and $3 \times 4 = 12$.

C1 · E-DIST

$6(x + 4) = 6x + 24$; 66 tools.

Reason: Each of six kits contributes x pencils and four markers. Thus $6(x + 4) = 6x + 24$. At $x = 7$, $42 + 24 = 66$. The expression $6x + 4$ counts markers from only one kit.

C2 · E-LIKE

$6x + 6$ square m; 30 square m; one 3 m by $(2x + 2)$ m rectangle.

Reason: The areas are $3x$ and $3(x + 2)$. Adding gives $6x + 6$. At $x = 4$ this is 30. Place the banners side by side: total width $2x + 2$, so $3(2x + 2)$ gives the same area.

Day 037 · Equations and related quantities

Workbook p. 47 · assigned set

R1 · P-PART

42 points

R2 · F-DIV

1 meters

R3 · D-DIV

40

R4 · R-TABLE

0, 12, and 21 centimeters

P1 · E-ADD

x = 12

Reason: Subtract 7 from each side.

P2 · E-MULT

4b = 28; b = 7 marbles

P3 · E-MULT

x = 4.5

Reason: Divide both sides by 1.2. Then 1.2 × 4.5 = 5.4.

P4 · E-MULT

x = 15

P5 · E-STORY

a + 6.50 = 18; a = $11.50

P6 · E-RELATE

(0, 0), (1, 2.5), (2, 5), (4, 10). Input n; output c.

Reason: Multiply each notebook count by 2.5. The chosen count determines the cost.

F · E-MULT

5; 7

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · E-MULT

x = 7. Six equal groups total 42.

Reason: Treats multiplication as addition. Draw six equal boxes with a total of 42.

M1 · P-WHOLE

80 shells

M2 · F-DIV

3

M3 · D-ADD

8.528

X1 · E-ADD

x = 3.

Reason: Subtract 4 from each side, then check by addition.

S · EXPLAIN

Answers vary. A solution makes the equation true. The same operation on both sides preserves equality.

Reason: Accept an accurate explanation supported by a matching example.

E · E-MULT

x = 20

Reason: Three fifths total 12, so one fifth is 4. Five fifths total 20.

C1 · E-STORY

$5 per bag; first shop is cheaper by $1 for five bags.

Reason: Remove the $4 fee: three bags cost $15, so each costs $5. Five from the first shop cost 5 × 5 + 4 = $29. Five from the second cost 5 × 6 = $30. The first shop saves $1.

C2 · E-RELATE

Table: (0, 3), (1, 5), (3, 9), (5, 13); L = 3 + 2t; 7 minutes.

Reason: The starting 3 liters remain at every time. Add 2t liters to get L = 3 + 2t. To reach 17, 14 more liters are needed; 14 ÷ 2 = 7 minutes. The rule 2t omits the initial water.

Day 038 · Expressions and equations

Workbook p. 49 · assigned set

1 · N-FACT

24

2 · N-FACT

56

3 · N-FACT

32

4 · N-FACT

72

5 · F-DIV

$1\frac{1}{3}$ meters

6 · F-ADD

$\frac{11}{12}$

7 · N-FACT

8

8 · E-EVAL

17

9 · E-DIST · review

2x + 14

10 · E-EXP · review

25

11 · E-EVAL · review

11

12 · E-EXPR · review

3b; 12 beads

13 · E-DIST

5z + 10. Each of the five groups contains two extra units.

14 · E-EXPR

5p + 4; 24 cards.

15 · E-EVAL

21

Reason: 4 × 4 + 5 = 16 + 5 = 21.

16 · E-EXP

21

Reason: The exponent gives two equal factors: 4 × 4; add 5 afterward.

17 · E-DIST

4x + 20; both equal 36.

Reason: Multiply both x and 5 by 4.

18 · E-LIKE

10y; 20.

C1 · E-DIST

4(2a + 12) = 8a + 48; $120; yes, $5 remains.

Reason: One pack costs 2a + 3 × 4 = 2a + 12. Four packs cost 8a + 48. Substituting a = 9 gives 72 + 48 = $120; 125 − 120 = $5 remains.

C2 · E-DIST

5(n + 2)−10 = 5n; 30 square m.

Reason: The whole garden has area 5n + 10. Removing 10 leaves 5n. For n = 6 the whole area is 40 and the path is 10, leaving 30. The added 2-meter strip has area 5 × 2 = 10.

Day 039 · Expressions and equations

Workbook p. 50 · assigned set

1 · F-EQUIV

0.5; 50%

2 · F-EQUIV

0.25; 25%

3 · F-EQUIV

0.75; 75%

4 · F-EQUIV

0.125; 12.5%

5 · F-DIV

$1\frac{3}{7}$

Reason: Keep four sevenths and multiply by five halves. The check recovers four sevenths.

6 · P-WHOLE

40 shells

7 · N-FACT

12

8 · E-MULT

x = 25

9 · E-ADD · review

x = 7

10 · E-MULT · review

x = 8

11 · E-MULT · review

x = 3.2

12 · E-MULT · review

x = 16

13 · E-MULT

x = 7. Six equal groups total 42.

14 · E-EXPR

6p + 5; 35 cards.

15 · E-EVAL

31

Reason: 5 × 5 + 6 = 25 + 6 = 31.

16 · E-EXP

31

Reason: The exponent gives two equal factors: 5 × 5; add 6 afterward.

17 · E-DIST

5x + 30; both equal 55.

Reason: Multiply both x and 6 by 5.

18 · E-LIKE

12y; 24.

C1 · E-STORY

15 m; L ÷ 4 − 1.5 = 2.25.

Reason: Undo the trim first: each original piece was 2.25 + 1.5 = 3.75 m. Four original pieces total 15 m. Check: 15 ÷ 4 − 1.5 = 3.75 − 1.5 = 2.25.

C2 · E-STORY

6 km; $5 each with six riders.

Reason: Five payments of $6 make a $30 bill. After the $18 starting fare, the distance charge is $12. At $2 per kilometer, the trip is 6 km. The unchanged $30 bill split six ways costs $5 each.

Day 040 · Expressions and equations

Workbook p. 51 · assigned set

1 · N-SCALE

82

2 · N-SCALE

8

3 · N-SCALE

17

4 · N-SCALE

24

5 · E-MULT

x = 3.2

6 · E-DIST

4x + 20

7 · D-MULT

0.0050

8 · N-DIV

12

9 · P-WHOLE

40 shells

10 · R-EQUIV

15 blue tiles

11 · N-FACT

24

12 · G-VOL

72 cm³

13 · E-STORY · review

6 and 11

Reason: Remove the extra 5: 17 − 5 = 12. Split the rest equally: 6 and 6. Restore 5 to one number.

14 · E-EXPR

7p + 2; 44 cards.

15 · E-EVAL

19

Reason: 2 × 6 + 7 = 12 + 7 = 19.

16 · E-EXP

11

Reason: The exponent gives two equal factors: 2 × 2; add 7 afterward.

17 · E-DIST

2x + 14; both equal 26.

Reason: Multiply both x and 7 by 2.

18 · E-LIKE

10y; 20.

19 · E-DIST

2(z + 7) = 2z + 14.

20 · E-ADD

x = 6.

Reason: Subtract 7 from each side, then check by addition.

21 · E-MULT

x = 6.

Reason: Divide both sides by 2; each of 2 equal groups is 6.

22 · E-MULT

x = 18.

23 · E-STORY

7p = 14; p = $2.

C1 · E-LIKE

8n − 2n + 9 = 6n + 9; n = $4.

Reason: Returning 2 of 8 notebooks leaves a net purchase of 6. Thus 6n + 9 = 33; notebooks account for 24 dollars, giving n = 4. Original cost 32, refund 8, pens 9: 32 − 8 + 9 = 33.

C2 · E-DIST

17 and 27; second is always 10 greater.

Reason: The first rule is 3x + 5. The second is 3(x + 5) = 3x + 15. At 4 the outputs are 17 and 27. Subtracting the first expression from the second leaves 10, independent of x.

Day 041 · Expressions and equations

Workbook p. 52 · assigned set

1 · N-EST

800

2 · N-EST

9

3 · N-EST

About 100

4 · N-EST

About 20

5 · E-MULT

x = 3.2

6 · E-DIST

4x + 20

7 · E-MULT

x = 8

8 · F-DIV

$2\frac{1}{2}$; stories vary

9 · D-DIV · review

1.2

10 · D-DIV · review

40

11 · D-MONEY · review

$2.57

12 · D-SUB · review

8.714

13 · E-EXPR

6b; 24 beads

14 · E-MULT

x = 7. Six equal groups total 42.

15 · E-EXPR

8p + 3; 59 cards.

16 · E-EVAL

29

Reason: 3 × 7 + 8 = 21 + 8 = 29.

17 · E-EXP

17

Reason: The exponent gives two equal factors: 3 × 3; add 8 afterward.

18 · E-DIST

3x + 24; both equal 45.

Reason: Multiply both x and 8 by 3.

C1 · E-STORY

Folder price $5.25; nine cost $47.25, so no; $1.50 short.

Reason: Remove delivery: $45.75 − $9 = $36.75 for seven folders, so each costs $5.25. Nine cost $47.25. Compared with the $45.75 budget, that is $1.50 too much.

C2 · E-EXP

A = x² + 2x; areas 3, 15, and 35 square cm; increases 12, then 20, so they are unequal.

Reason: At x = 3, 9 + 6 = 15. At x = 5, 25 + 10 = 35, an increase of 20. For comparison x = 1 gives 3, so increasing from 1 to 3 adds 12. The square term makes equal length changes produce different area changes.

Day 042 · Weekly check

Workbook p. 53 · assigned set

1 · E-MULT

x = 3.2

2 · E-DIST

4x + 20

3 · E-MULT

x = 8

4 · E-EVAL

13

5 · D-SUB

8.714

6 · D-MULT

0.0100

7 · D-MULT

0.0050

8 · P-WHOLE

40 shells

9 · R-EQUIV

15 blue tiles

10 · E-DIST

5z + 10. Each of the five groups contains two extra units.

11 · E-EXPR

9p + 4; 76 cards.

12 · E-EVAL

41

Reason: 4 × 8 + 9 = 32 + 9 = 41.

13 · E-EXP

25

Reason: The exponent gives two equal factors: 4 × 4; add 9 afterward.

14 · E-DIST

4x + 36; both equal 68.

Reason: Multiply both x and 9 by 4.

15 · E-LIKE

14y; 28.

16 · E-DIST

4(z + 9) = 4z + 36.

17 · E-ADD

x = 8.

Reason: Subtract 9 from each side, then check by addition.

18 · E-MULT

x = 8.

Reason: Divide both sides by 4; each of 4 equal groups is 8.

C1 · E-RELATE

For 4: A $24, B $20; for 8: A $36, B $40; equal at 6 bracelets, $30.

Reason: A has a $12 head start but grows $2 less per bracelet than B. Six bracelets close that $12 gap. Check 12 + 3 × 6 = 5 × 6 = 30. The cheaper option changes between 4 and 8.

C2 · E-DIST

Correct form 5x + 10; at 2, 20≠14; they agree only at x = 5.

Reason: Distribute first: 2x + 10 + 3x = 5x + 10. At x = 2 this is 20, while 7x = 14. Agreement requires the extra 2x in 7x to equal 10, so x = 5. One matching input does not prove equivalence.